

Name: _____ Partner: _____ Period: _____

Mission Control — Parametrizing Conic Paths

Mission Control is tracking three objects. Each follows a conic section path described by an implicit equation. Your job: write a parametrization for each object, then use parametric tools to extract key position data.

Object 1 — Space Station (Ellipse) — Key

The station's orbital path satisfies $\frac{(x-1)^2}{64} + \frac{(y+2)^2}{36} = 1$.

(a) $h = 1$ $k = -2$ $a = 8$ $b = 6$

(b) Parametrization:

$x(t) = 1 + 8 \cos t$ $y(t) = -2 + 6 \sin t$

(c) Four extreme positions:

Rightmost (x max): $t = 0$ Point: $(9, -2)$

Leftmost (x min): $t = \pi$ Point: $(-7, -2)$

Highest (y max): $t = \pi/2$ Point: $(1, 4)$

Lowest (y min): $t = 3\pi/2$ Point: $(1, -8)$

(d) y -axis crossing: set $x(t) = 0 \Rightarrow 1 + 8 \cos t = 0 \Rightarrow \cos t = -\frac{1}{8}$

$t = \arccos(-\frac{1}{8}) \approx 1.696$ **rad** (and $2\pi - 1.696 \approx 4.587$ **rad**)

y -axis crossing coordinates: $(0, -2 + 6 \sin(\arccos(-1/8))) \approx (0, 3.97)$ **and** $(0, -7.97)$

Teacher Note

Students often stop at $\cos t = -1/8$ and forget to find the y -coordinate. Prompt: "You found the time t — now plug it back into $y(t)$ to get the position." For the exact form: $\sin(\arccos(-1/8)) = \pm\sqrt{1 - 1/64} = \pm\sqrt{63}/8 = \pm3\sqrt{7}/8$.

Object 2 — Comet (Parabola) — Key

(a) Vertex: $(1, -3)$ Orientation: **vertical (opens upward)**

(b) $x(t) = 1 + t$, $y(t) = 2t^2 - 3$

(c) x -axis: set $y(t) = 2t^2 - 3 = 0$:

$t = \pm\sqrt{3/2} = \pm\frac{\sqrt{6}}{2} \approx \pm 1.225$

$x = 1 + t$: $x = 1 + \frac{\sqrt{6}}{2} \approx 2.225$ **and** $x = 1 - \frac{\sqrt{6}}{2} \approx -0.225$

Coordinates: $(1 + \frac{\sqrt{6}}{2}, 0)$ **and** $(1 - \frac{\sqrt{6}}{2}, 0)$

Object 3 — Satellite (Circle) — Key

(a) Center: $(0,0)$ Radius: 10

(b) $x(t) = 10 \cos t$ $y(t) = 10 \sin t$

Domain: $0 \leq t \leq 2\pi$

(c) At $t = \pi/4$:

$$x(\pi/4) = 10 \cos(\pi/4) = 10 \cdot \frac{\sqrt{2}}{2} = 5\sqrt{2} \approx 7.07$$

$$y(\pi/4) = 10 \sin(\pi/4) = 10 \cdot \frac{\sqrt{2}}{2} = 5\sqrt{2} \approx 7.07$$

Coordinates: $(5\sqrt{2}, 5\sqrt{2})$

Teacher Note

A common error: students write $r = 100$ (confusing r^2 with r). Remind them: $x^2 + y^2 = r^2$, so $r^2 = 100$ gives $r = 10$.